TASK 4

Case Classification & Severity Analysis

Pharmacovigilance Training Report | Simulated Case Review

Objective
Classify adverse drug reaction (ADR) cases by seriousness and expectedness, assess potential clinical impact, and recommend appropriate documentation and follow-up actions.

Scope and data source
This report uses five simulated educational cases corresponding to the medicines and reactions shown in the task screenshot: amoxicillin, ibuprofen, metformin, atorvastatin and levofloxacin. It contains no real patient data.

Important interpretation note
Expectedness labels below are training assumptions and must be checked against the current local product information or applicable reference safety information before any real-world reporting decision. The report is educational and does not replace clinical judgement or local regulatory requirements.

1. Methodology

Seriousness: A case is generally considered serious if it results in death, is life-threatening, requires or prolongs inpatient hospitalization, causes persistent/significant disability, results in a congenital anomaly, or qualifies as another medically important event. A symptom can be intense without meeting a seriousness criterion; seriousness is assessed by outcome and medical significance.

Expectedness: Compare the reaction's nature and severity with the current applicable product label/reference safety information. If the reaction is not listed, or differs materially in specificity or severity, it may be unexpected. The applicable reference document depends on the reporting context.

Clinical impact: Summarize plausible risk and what information should be documented, without inferring causality from temporal association alone.

2. Case Summary

Case	Drug	Adverse reaction	Seriousness	Expectedness
01	Amoxicillin	Skin rash and itching	Non-serious	Expected*
02	Ibuprofen	Gastrointestinal bleeding requiring hospitalization	Serious	Expected*
03	Metformin	Lactic acidosis	Serious	Unexpected*
04	Atorvastatin	Mild elevation of liver enzymes	Non-serious	Expected*
05	Levofloxacin	Tendon rupture	Serious	Expected*

Overall result: 5 simulated cases reviewed — 3 classified as serious and 2 as non-serious. Expectedness assumptions: 4 expected and 1 unexpected. Counts reflect the illustrative dataset only.

3. Case-by-Case Clinical Impact & Recommendations

Case 01 — Amoxicillin

Reaction: Skin rash and itching
Classification: Non-serious | **Expectedness:** Expected*

Potential clinical impact: Mild cutaneous symptoms; monitor for progression or signs of systemic allergy.

Recommended follow-up: Document onset and severity; assess allergy history; seek urgent care if breathing difficulty, facial swelling, blistering, or rapid worsening occurs.

Case 02 — Ibuprofen

Reaction: Gastrointestinal bleeding requiring hospitalization
Classification: Serious | **Expectedness:** Expected*

Potential clinical impact: Potentially major blood loss and hemodynamic compromise; hospitalization is a seriousness criterion.

Recommended follow-up: Urgent clinical management; document admission and outcome; review concomitant medicines and risk factors; report through the applicable pharmacovigilance pathway.

Case 03 — Metformin

Reaction: Lactic acidosis
Classification: Serious | **Expectedness:** Unexpected*

Potential clinical impact: Potentially life-threatening metabolic emergency requiring prompt assessment and treatment.

Recommended follow-up: Urgent medical assessment; record laboratory evidence, renal function, contributing conditions and outcome; verify expectedness against the current product information.

Case 04 — Atorvastatin

Reaction: Mild elevation of liver enzymes
Classification: Non-serious | **Expectedness:** Expected*

Potential clinical impact: Laboratory abnormality described as mild in this training scenario; clinical significance depends on magnitude, symptoms and trend.

Recommended follow-up: Document values and dates; assess symptoms and trend; clinician to decide monitoring or treatment changes according to current guidance and product information.

Case 05 — Levofloxacin

Reaction: Tendon rupture
Classification: Serious | **Expectedness:** Expected*

Potential clinical impact: May cause substantial functional impairment and require specialist treatment or surgery; assess seriousness based on actual outcome.

Recommended follow-up: Prompt clinical evaluation and documentation of functional impact; record onset, treatment, outcome and relevant risk factors; verify local product label and reporting requirements.

---

4. Key Findings

• Serious cases in this scenario are ibuprofen-associated gastrointestinal bleeding requiring hospitalization, metformin-associated lactic acidosis, and levofloxacin-associated tendon rupture. Hospitalization is a seriousness criterion; lactic acidosis may be life-threatening; tendon rupture can lead to significant functional impairment. The actual case outcome and clinical facts determine final classification.

• The non-serious examples are a skin rash/itching case without reported serious outcome and a mild liver-enzyme elevation without reported clinical complications. Reassess if the reaction progresses, requires hospitalization, or meets another seriousness criterion.

• Expectedness is separate from seriousness. A known/labelled reaction can still be serious, and a non-serious event can be unexpected. Verify the reaction and its severity against the exact product information relevant to the case.

5. General Recommendations

• Capture the minimum case details: identifiable reporter and patient (handled confidentially), suspect product, reaction/event, and a reportable date or timeframe, following local requirements.

• Record dose, route, indication, start/stop dates, time to onset, relevant medical history, concomitant medicines, laboratory findings, treatment, dechallenge/rechallenge information when available, and patient outcome.

• For serious events, document the seriousness criterion explicitly (for example, initial hospitalization) and follow applicable expedited reporting timelines and internal escalation procedures.

• Do not treat a suspected association as proof of causality. Record evidence and uncertainty objectively; request follow-up information where appropriate.

• Protect privacy: use fictional or de-identified training data in a public GitHub repository. Never upload identifiable patient details, medical records, or confidential company information.

6. Conclusion

The simulated review demonstrates that seriousness and expectedness are distinct pharmacovigilance assessments. Consistent case documentation, checking current product information, documenting clinical outcomes, and following local reporting procedures are essential to reliable safety surveillance. The classifications in this report are educational examples and should not be interpreted as formal medical or regulatory determinations.